Associations between maternal phthalate exposure and cord sex hormones in human infants.

It has been speculated that maternal phthalate exposure may affect reproductive development in human newborns. However, the mechanism awaits further investigation. The aim is to evaluate the association between maternal phthalate exposure and cord sex steroid hormones in pregnant women and their newborns from the general population. A total of 155 maternal and infant pair were recruited and analyzed. Levels of urinary phthalate metabolites and sex steroid hormones were determined using liquid chromatography/electrospray tandem mass spectrometry (LC-ESI-MS/MS) and radioimmunoassay (RIA), respectively. No significant correlation was found between each steroid hormones and phthalate metabolites for male newborns, except MMP was marginally significantly correlated with E(2). After adjusting for maternal age, estradiol (E(2)) levels in cord serum from male newborns were not correlated with maternal urinary phthalate metabolites. In female newborns, the maternal urinary levels of mono-(2-ethylhexyl) phthalate (MEHP) and mono-(2-ethyl-5-hydroxyhexyl) phthalate (5OH-MEHP) were negatively correlated with the free testosterone (fT) and fT/E(2) levels in cord serum with Pearson correlation coefficients ranging between -0.24 and -0.29 ($p<0.05$). Additionally, after gestational age was adjusted, the maternal urinary level of DEHP was negatively correlated with the free testosterone (fT) and fT/E(2) levels in cord serum. We suggest that maternal exposure to phthalates may affect sex steroid hormones status in fetal and newborn stage.